ARAGOCOR MINERALS

Oolitic Aragonite (Unprocessed) — Technical Data

High-purity natural oolitic aragonite · Technical documentation

Technical Data

Oolitic Aragonite (unprocessed)

Oolitic aragonite - the world's only recent origin, renewable and sustainable performance mineral

Oolitic Aragonite (unprocessed)

Exceptionally Pure calcium carbonate

Oolitic Aragonite, a naturally precipitated high surface area calcium carbonate in the crystalline form of aragonite. This lower carbon footprint natural material can be used in place of limestone derived products. The higher solubility of aragonite versus calcite in combination with an extremely large surface area associated with oolites and an extremely pure material (>39% calcium) provides an extremely efficient calcium carbonate.

The solubility of aragonite is determined by its unique crystalline structure. Acid neutralization curves for aragonite were developed by the Keck Geophysical Laboratory at Stevens Institute of Technology and compared with an equilibrium curve developed using Visual Minteq Chemical Modeling Software (KTH, 2010) for Calcite (Figure 1). Visual Minteq uses the process and chemical databases provided by the USEPA model MINTEQA2 (USEPA, 2010). Results clearly indicate that use of aragonite supplies greater concentrations of calcium over a pH range of 5-7. This result implies that at a pH of 5.5 it will require far less aragonite than calcite to deliver the same dose of calcium. The aragonite measurements were completed on unaltered oolites with a particle size in the range of 75-850 microns and have been compared to an idealized, infinitely small particle size, 100% pure calcite. The surface area of the oolitic aragonite has been measured in excess of 1.7 m2/g (approximately 124,667 in2/in3).

Aragonite Composition Large surface area and porosity

Properties

Oolitic Aragonite

Calcite/Limestone

Specific Gravity

2.93+

2.7

Mohs Hardness

3.5 - 4

3

Crystal Structure

orthorhombic

hexagonal

Surface Area

> 1.82m2/g

< .55m2/g

Zeta Potential

-33.85mV to - 6.65mV

-1.01mV to +11.55Mv

Micro porosity

Very high

Low

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